MILITARY SPECIFICATION

TEST SET, PRESSURE INDICATING SYSTEM TTU-26/E, VARIABLE RELUCTANCE, FIELD

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of pressure indicating system test set designated TTU-26/E.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225	Aluminum Alloy Bar, Rod, Wire or Special Shapes, Rolled, Drawn. or Cold Finished, General Specification for.
QQ-A-250	Aluminum and Aluminum Alloy Plate and Sheet, General Specification for.
QQ-S-571	Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Box, Fiberboard.

MILITARY

MIL-P-116	Preservation, Methods of.
DOD-D-1000	Drawings, Engineering and Associated Lists.
MIL-C-5015	Connectors, Electric, AN Type, General Specification for.
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-7742	Screw Thread, Standard, Optimum Selected Series, General Specification for.
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-W-16878	Wire, Electrical, Insulated High Temperature.
MIL-T-25435	Transmitters, Pressure Aircraft, Variable Reluctance Type, General Specification for.
MIL-I-25437	Indicators, Aircraft, Pressure, Voltage Radio Type, General Specification for.
MIL-I-25438	Indicator, Pressure, Type MO-2.
MIL-I-38230	Transmitter, Pressure TRU-66/A Oil, Variable Reluctance.
MIL-S-45743	Soldering, Manual Type, High Reliability, Electrical and Electrical Equipment.

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications Order of Precedence for.
MIL-STD-147	Palletized and Containerized Unit Loads, 40 Inch X 48 Pallets, Skids, Runners or Pallet Type Base.
MIL-STD-202	Test Methods for Electronic and Electrical component Parts.
MIL-STD-794	Part and Equipment Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods & Engineering Guidelines.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1516	Coating for Aircraft and Missiles.
MS3101	Connector, Plug, Electric, Cable Connecting, Solder Contacts, AN Type.
MS3102	Connector, Receptacle, Electric, Cable Connecting, Solder Contacts, AN Type.
MS3106	Connector, Plug, Electric Straight, Solder Contacts, AN Type.
MS24524	Switch, Toggle, Two Pole, Environmentally Sealed.
MS25043	Cover, Electrical Connector Receptacle, AN Type.
MS25307	Switch, Toggle, Positive Break, Environmentally Sealed, Screw Terminal, Double Pole, 469 Mounting Bushing, 25 Amperes.
MS28042	Clamp, Instrument Mounting, Aircraft.
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standing Practice for Commercial Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. This specification makes provision for first article testing.

3.2 Component parts. The test set shall consist of a single integral unit incorporating a carrying case, a master indicator, a variable ratio transformer, a frequency meter, a voltmeter, a variable transformer for voltage adjustment, and associated switches and adapter cables.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts such as screws, bolts, nuts, et cetera, having suitable properties, may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.3.2 Standard parts. With the exception of the preceding paragraph, AN and MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are to be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

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3.4.2 Protective treatment. When materials are used in the construction of the tester that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.3 Materials shall conform to applicable specifications as specified herein. When materials are used that are not specifically designated, they shall be entirely suitable for the purpose.

3.4.4 Metals. Metals shall be of the corrosion-resistant type unless suitably protected to resist corrosion during storage and normal service use.

3.4.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined by MIL-STD-889.

3.5 Design and construction.

* 3.5.1 The test set shall be designed as a portable unit suitable for testing variable reluctance-type pressure transmitters and voltage ratio-type pressure indicators as covered by MIL-T-25435, MIL-T-38230, MIL-I-25437 and MIL-I-25438. A precision voltage reference (variable ratio transformer) shall be incorporated for testing transmitters. Means shall be provided for checking the master indicator against the precision voltage reference. Provision shall also be incorporated for facilitating checking of both transmitters and indicators in aircraft, using aircraft power.

3.5.2 General layout. The general layout of the test set shall conform substantially to figures 1 and 2. The detailed mechanical and electrical design of the test set shall be accomplished by the contractor subject only to the requirements of this specification, these requirements being detailed only to the extent considered necessary to obtain desired mechanical and electrical characteristics, performance, and permanence. The design, layout, and assembly of the test set and its component parts shall facilitate quantity production and ease of maintenance and operation. The component parts incorporated in the test set shall be electrically connected as shown on figure 3.

3.5.3 The construction of the test set shall be mechanically and electrically sound, suitable for the purpose intended, and give assurance of permanence in accuracy of indication and output. The design and construction of the test set shall take into account the temperature, vibration, shock, humidity, and operational requirements specified herein.

3.6 Performance. The test set shall be capable of meeting the tests specified herein when subjected to the following conditions:

a. Temperature - Temperatures ranging from -62°C to +71°C.

b. Vibration - Virbration at frequencies varying between 10 and 55 cycles per second at a total excursion of 0.03 ±0.006 inch for a period of 1 hour in each of three mutually perpendicular axes.

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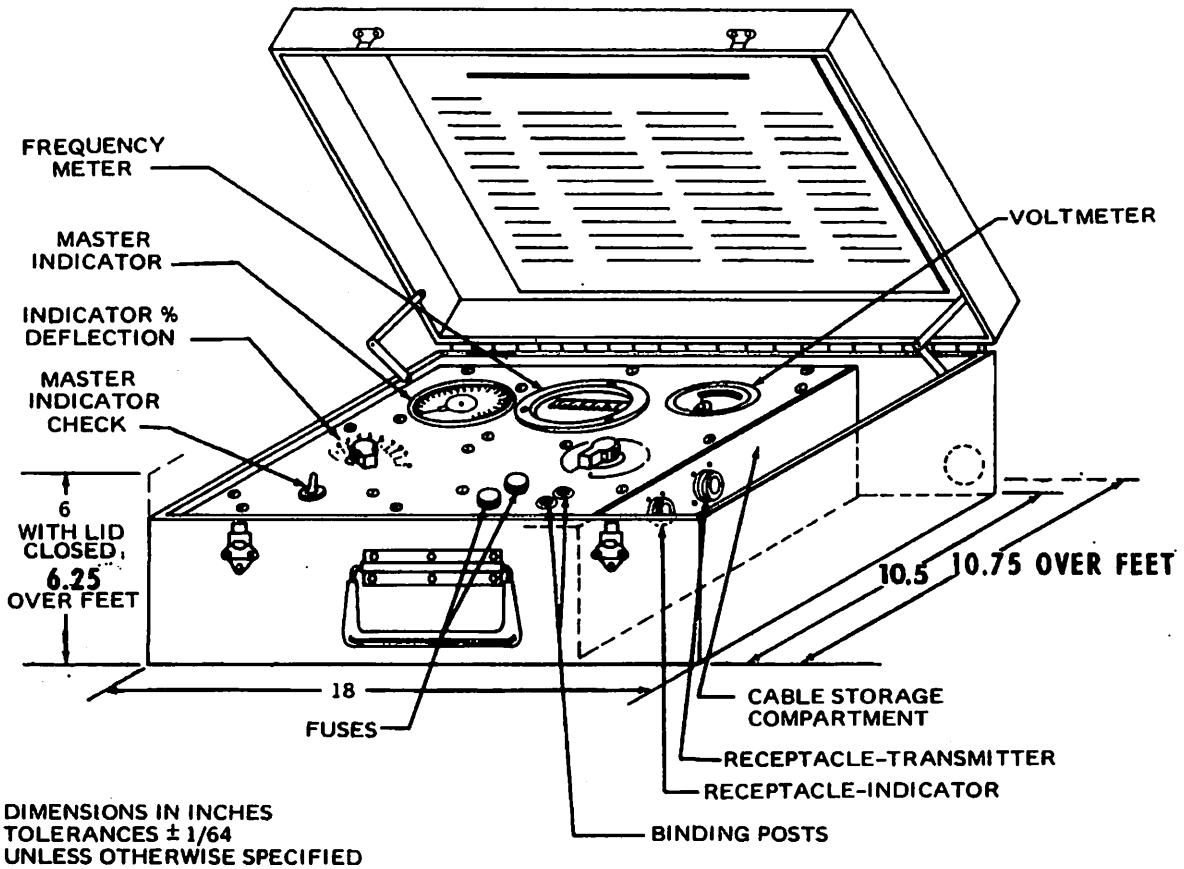


FIGURE 1.

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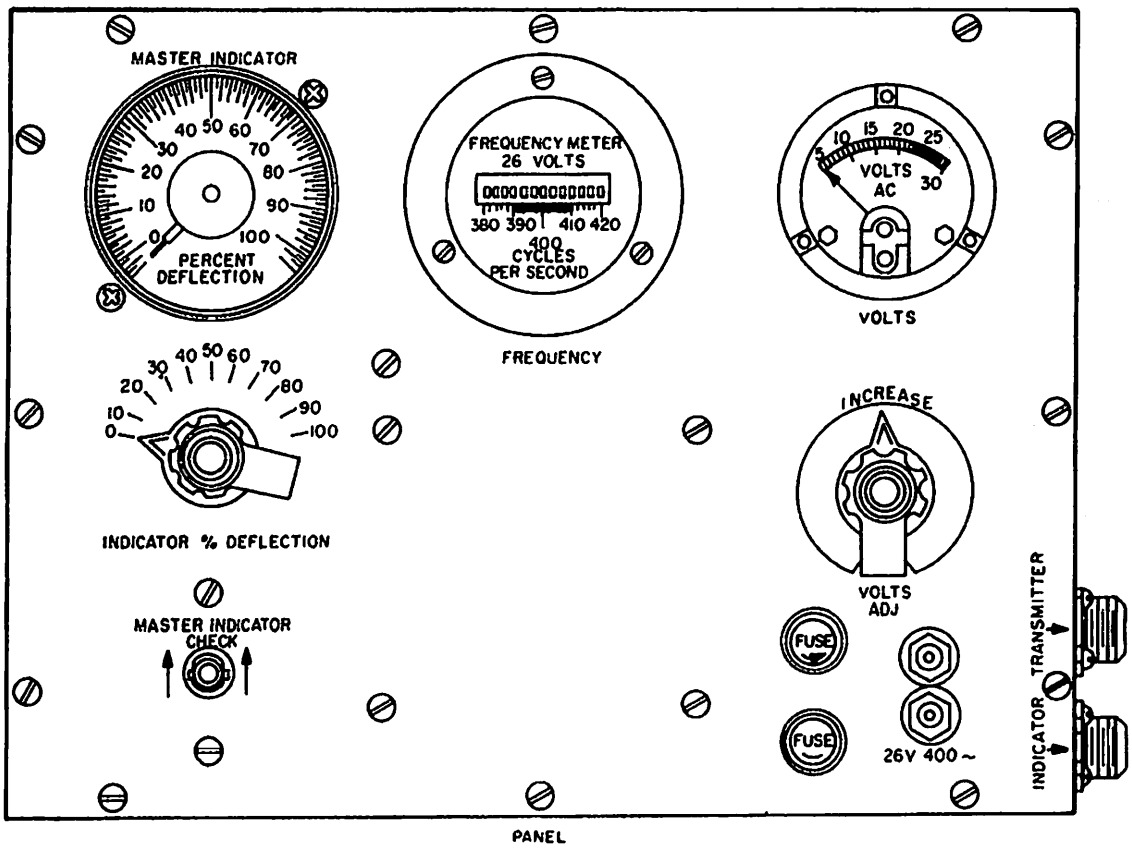


FIGURE 2.

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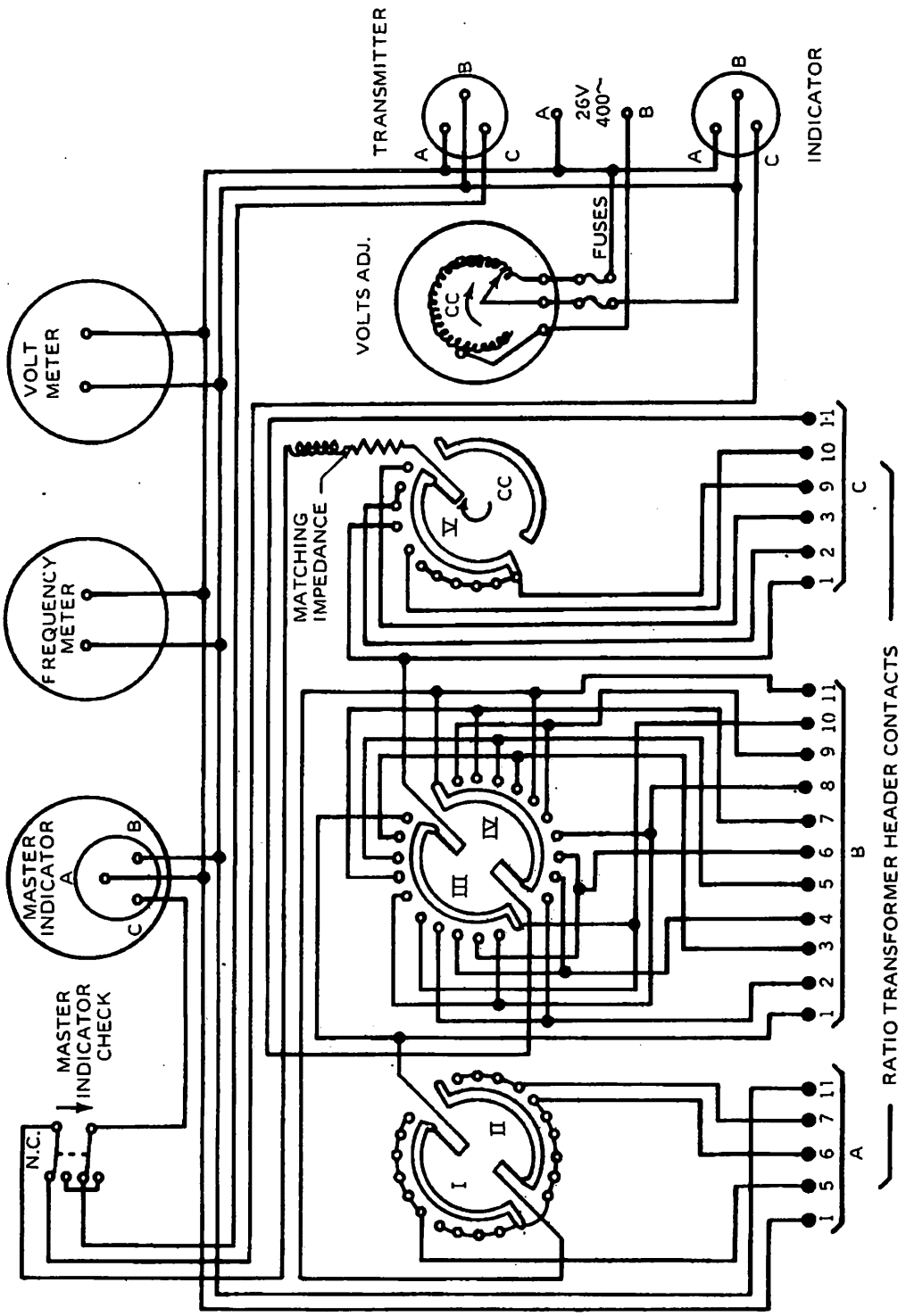


FIGURE 3.

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c. Shock - Impact shocks of 20 gravity (g).

d. Humidity - Relative humidity up to 95 percent for a period of 240 hours (10 cycles).

* 3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in manufacturer's part number.

3.8 Case. The tester shall be enclosed in a carrying case, as shown on figure 1, and be designed to facilitate inspection and maintenance of component parts incorporated therein.

* 3.8.1 Case material. The case, cover, and control panel shall be fabricated from aluminum alloy conforming to QQ-A-250. Material required for reinforcement purposes shall conform to QQ-A-225 as applicable. The thickness of the material shall be a minimum of 0.081 inch for the case, 0.064 inch for the cover, and 0.125 inch for the panel. The case shall be sturdy.

* 3.8.2 Case finish. The inside and outside of the case including the panel shall be anodized per MIL-A-8625, and shall be finished in accordance with MIL-STD-1516, color 13538 as specified in FED-STD-595.

3.8.3 Dimensions. The outside dimensions of the case, including projections introduced by the latches, feet, and handle when folded down, shall be maintained with a maximum length of 18 inches, a width of 12 inches, and a depth of 7 inches.

3.8.4 Feet. The case shall be provided with 4 feet on the hinged side of the case and 4 feet on the side opposite the lid. The feet shall be formed by dimpling or other suitable means.

3.8.5 Handle. A hinged metal handle shall be mounted on the outside surface of the case, as shown on figure 1, and shall be located in the optimum position with respect to weight distribution. The handle shall provide clearance from the case sufficient to pass a block 1.875 by 4.375 in cross section with the edges rounded to a .9375 inch radius. The grip portion of the handle shall be of nonmetallic material, shaped to fit the hand comfortably.

3.8.6 Lid. The case shall be provided with a nondetachable lid, secured to the case by a hinge of the continuous type. Two sturdy lid supports fabricated from corrosion resistant steel shall be provided, as shown on figure 1, for relieving hinge strain when the lid is open. Provisions shall be incorporated at the case and cover joints to keep the sides of the lid rigid and flush with the respective sides of the case when the lid is closed and the latches secured.

3.8.7 Storage compartment. A compartment shall be provided in the case as shown on figure 1 for storage of connector cables. The width of the compartment shall be 4 $\pm$.25 inch.

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3.8.8 Latches. The case shall be provided with sturdy latches of the linklock type. The latches shall be located as shown on figure 1 and shall conform to Simmons Fastener Corporation No. SL-3, or equal. The latches shall be fabricated of corrosion-resistant steel.

3.9 Controls. All controls provided for the test set shall be readily accessible, suitably arranged, and of such size and construction to permit convenience and ease of operation by operating personnel under all applicable service conditions. The controls shall rotate freely and smoothly without binding or excessive lost motion. The controls shall be adequately lubricated where necessary.

3.9.1 Control panel. The case shall incorporate a control panel as shown on Figure 1 for mounting the voltmeter, variable voltage control, frequency meter, master indicator, variable ratio transformer, associated switches, and receptacles. The panel layout shall be as shown on Figure 2.

3.9.1.1 Control markings. The test and approximate location of markings for the control panel shall be as shown on Figure 2. The markings shall be engraved in the panel and shall be 0.125 $\pm$0.010 inch in height and 0.020 $\pm$0.005 inch in width. The depth of the markings shall be 0.015 $\pm$0.005 inch. Markings shall be filled with black enamel or other suitable material.

3.9.2 Control knobs. All knobs shown on Figure 2 shall be of one-piece molded plastic, black, provided with a bored brass insert and bushing for a 1/4-inch shaft, and be firmly secured to their respective control shafts by a suitable retaining device. The knobs shall incorporate a bar-type extension to facilitate operation. The skirt diameter of the knob shall be approximately 1.3 inch. The pointer shall be provided with a white reference marking.

3.10 Master indicator. The master indicator specified for the test set shall incorporate an electrical movement of the voltage ratio type as covered in MIL-I-25437 and MIL-I-25438. It shall conform to Armtec Industries Inc., Instrument Division Part No. 289-100A, or equal. It shall be secured to the control panel by means of an instrument clamp conforming to MS28042-2.

3.10.1 Case. The master indicator shall be provided with a 3-1/4 inch size instrument case conforming to MS33639. The overall length of the case including the protrusion of the connector receptacle shall not exceed 3 inches. The case shall be constructed of suitable material applicable to hermetic sealing, uniform in texture with a smooth external surface. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. Means shall be provided for removal and replacement of the internal mechanism from the case and the case resealed, without the use of special tools and fixtures unless approved by the procuring activity. The hermetic sealing shall be such that the seal will not be dependent upon materials that will be affected by any atmosphere to which the indicator may be subjected.

3.10.1.1 The case specified for the master indicator shall be filled with a medium that is at least 98 percent pure, free of dust particles, and contains no more than 0.006 milligram of water vapor per liter (dewpoint - 65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen with the remainder of helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. Where practicable, the 100 percent helium-filling medium shall be used.

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3.10.1.2 Cover glass. The cover glass provided for the master indicator case shall be suitable for hermetic sealing and shall be clear, flat, and free from flaws that interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose intended.

3.10.2 Dial. The dial provided for the master indicator shall be made preferably of aluminum and shall be fastened securely to the case or upon the frame of the mechanism by not less than two screws in such manner that they will not loosen or turn when the instrument is vibrated.

3.10.2.1 Dial to cover glass distance. The distance between the inside surface of the cover glass and the dial shall not exceed 0.125 inch.

3.10.2.2 Dial scale. The dial scale provided for the master indicator shall be graduated and numbered to indicate percent deflection. The dial layout shall conform to Figure 2. The angular degree span of the scale from 0 to 100 percent shall not be less than 250 degrees. Major, intermediate, and minor graduations shall be provided at even intervals of 10, 1, and 0.5 percent deflection.

3.10.2.3 Dial markings. All markings shall be durable to withstand usage encountered in service. The graduation markings shall be linear within ±0 degrees 20 minutes of arc. The marking dimensions shall be in accordance with Table I.

TABLE I - DIAL MARKINGS

Markings	Height or length inch ±0.010	Width of line or graduation inch ±0.005	Material or finish
Numerals	0.156	0.031	Durable dull black
Percent deflection	0.156	0.031	Durable dull black
Major graduations	0.312	0.015	Durable dull black
Intermediate graduations	0.187	0.010	Durable dull black
Minor Graduations	0.125	0.010	Durable dull black
Background of dial	---	---	Durable dull white

3.10.3 Pointer. The pointer provided for the master indicator shall be light and sufficiently rigid to prevent oscillation under vibration. It shall be finished in durable dull black and be firmly attached to the associated mechanism. The pointer length shall allow the pointer tip to extend into the scale equal to 1/3 to 2/3 of the length of the shortest graduation. A knife-edge tip shall be provided for the pointer.

3.11 Voltmeter. An alternating-current voltmeter (400 cycles per second, nominal frequency) conforming to MIL-M-10304 shall be provided on the control panel, as shown on Figure 2. It shall provide a full-scale range of 30 volts. The case shall be of the 2.5 inch size. An accuracy of ±2 percent of full scale shall apply. The dial marking shall be finished in durable dull black on a durable dull white dial background. The dial layout shall be in accordance with Figure 2 with the portion of the dial from 24 to 28 volts shaded in green.

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3.12 Frequency meter. A ruggedized resonant reed-type frequency meter shall be provided on the control panel, as shown on Figure 2. The indicating range shall be from 380 to 420 cycles per second. It shall be designed for use with a 26-volt alternating-current power source. The dial aperture shall be approximately 2.5 inches. The case shall be hermetically sealed. The dial markings shall be finished in durable dull black on a durable dull white dial background. The dial layout shall be in accordance with Figure 2 with the portion of the dial from 390 to 410 cycles per second (cps) shaded in green. The frequency meter shall conform to H.H. Sticht Company Part No. 367-S2, or equal, and have an accuracy of $\pm$0.5 percent of rated frequency.

3.13 Variable power transformers. The variable power transformer required for adjustment of alternating-current input power shall be in accordance with Superior Electric Company Part No. 51070, or equal. The design shall permit an output of 26 volts when the input voltage varies over the range of 20 to 32 volts. The variable power transformer shall have a control knob for adjustment, mounted on the control panel in the position shown on Figure 2.

3.14 Variable ratio transformer. The variable ratio transformer specified for the test set shall conform to Gertsch Products, Inc., Model 246, or equal. It shall be designed to operate from a 26 volts, 400 cps, single phase, input power source. The accuracy of the unloaded transformer shall be $\pm\left(0.001 \text{ percent } \frac{-0.0001 \text{ percent}}{R}\right)$ where R is the indicated ratio. The maximum effective series inductance shall be 75 microhenries. The maximum effective series resistance shall be 2.0 ohms.

* 3.15 Switches. The basic design of the rotary type switch shown on Figure 2 (INDICATOR % DEFLECTION) shall incorporate a positive mechanical index locating each contact position. The index mechanism shall be designed to prevent the movable element from coming to rest between contact positions. The material used for the contacts and the processing of the material shall be suitable for the purpose intended. Stops shall be provided to limit travel of the switches beyond the design range and be sufficiently rugged to prevent damage to the mechanism. The switch provided for the MASTER INDICATOR CHECK circuit shall conform to Honeywell, Inc., Micro Switch Division, PN 532TS1-8 or Eaton Corp. Operations and Technical Center PN 8715K15 or equal. The power ON-OFF switch shall conform to MS24524 and MS25307.

3.16 Electrical connectors. Unless otherwise specified, electrical connectors shall conform to MIL-C-5015.

3.17 Internal wiring. All electrical connections within the test set shall be supported in order to alleviate breakage and minimize changes in measuring circuitry caused by strains, jars, vibrations, and other conditions incident to shipping, storage, and service. Soldering lugs or terminals shall be provided for all points of connection.

3.17.1 Hookup wire required for connecting the component parts of the test set shall conform to MIL-W-16878. The wiring shall be grouped together wherever practicable and shall be adequately secured within the test set.

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3.17.2 Two connector receptacles conforming to MS3102-10SL-3P shall be provided as shown on Figure 2. They shall be identified respectively by TRANSMITTER and INDICATOR markings on the panel. Screw-top binding posts conforming to Entran Inc. Part No. 640, or equal, shall be mounted on the control panel as shown on figure 2 for facilitating the connection of external power.

3.18 Adapter cables. Two adapter cables shall be provided for the test set as shown on Figure 4. The connectors provided for one end of the cables shall mate with the receptacles provided on the test set. The opposite end shall mate with transmitters or indicators being tested. In addition, an extension shall be provided on the adapter cables as shown on Figure 4. The extension is required for permitting input power to be obtained from the aircraft through the connector plug which has been removed from the instrument undergoing test. The adapter cables shall be wired A to A, B to B, and C to C. A and B in the extension connector plug shall connect to the A and B leads respectively in the basic adapter cable. The C connector pin shall not be used. The overall length of the adapter cables shall be 8 feet ±1 inch. The overall length of the extension shall be 15 ±1/2 inch as measured between the end of the adjacent connector provided on the adapter cable and the end of the connector plug provided on the extension.

3.19 Soldering. Soldering shall conform to MIL-S-45743. Solder used for the electrical connections shall be in accordance with QQ-S-571 and of suitable composition.

3.19.1 Soldering flux. Only non-activated rosin and alcohol, or the equivalent plastic rosin mixture shall be used as a flux in the assembly of electrical wiring and adapter cables.

3.20 Weight. The weight of the test shall not exceed 18 pounds.

3.21 Wiring diagram. A permanent wiring diagram as large in size as practicable shall be mounted on the inside bottom surface of the case. The diagram shall contain pertinent information on the electrical system to which reference can be made during servicing operations.

3.22 Operating instructions. Operating instructions shall be attached to the inside surface of the case lid as shown on Figure 1. The body of the chart shall be white and the markings black. The operating instructions shall be legible and capable of withstanding the effects of the tests specified in Section 4. The operating instructions shall concisely outline the procedures to follow in using the test set.

3.23 Finishes and protective coatings.

3.23.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

3.23.1.1 Dials, small holes and inserts need not be anodized.

3.23.1.2 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

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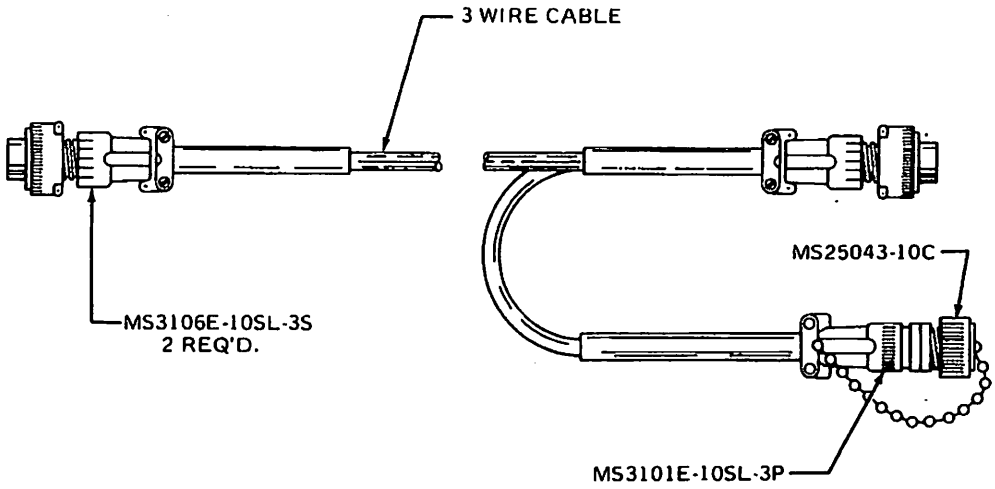


FIGURE 4.

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3.23.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in place of anodizing.

3.23.1.4 Where abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.24 Screw threads. Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter shall conform to MIL-S-7742.

3.25 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.26 Workmanship. The test set including all parts and accessories shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.26.1 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.26.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.26.3 Cleaning. The test set shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

3.26.4 Dimensions. The dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the tester, they shall be held or limited accordingly.

3.27 Reclaimed Materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the test set shall be classified as follows:

a. First article tests.

b. Quality conformance inspections.

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4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C) with the relative humidity not exceeding 60 percent. When tests are made with atmospheric pressure, humidity, or room temperature differing materially from the above specified values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Position. Unless otherwise specified, the test set shall be tested in its normal operating position.

4.3.3 Test power. Unless otherwise specified herein, the test set shall be tested with 26 volts, 400 cps alternating current. The voltage tolerance shall be ± 4 volts and the frequency tolerance shall ± 2 cycles.

4.3.3.1 Prior to taking any readings, power shall be applied to the test set for at least 3 minutes. Immediately after the warmup period, the voltage shall be adjusted by means of the VOLTS ADJUST control to give a reading on the voltmeter of 26 ± 1 volt.

4.3.4 Master test ratio transformer. A master test ratio transformer shall be used for checking the installed ratio transformer. Testing shall be on a comparison basis. The accuracy of the item shall be established and shall not exceed $\pm\left(.001 \text{ percent} + \frac{.0001 \text{ percent}}{R}\right)$ where R is the indicated ratio.

* 4.4 First article testing.

4.4.1 Test sample. The test sample shall consist of two test sets representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. A copy of a test report in accordance with MIL-STD-831 shall be submitted to the procuring activity.

* 4.4.3 First article tests. The first article tests shall consist of all tests described under 4.6.

4.5 Quality conformance inspection. Quality conformance inspection tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each test set shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Room temperature operation

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4.5.2 Sampling plans and tests.

4.5.2.1 Lot. Inspection lots shall be formed as specified in MIL-STD-105 from units that have passed the individual test.

4.5.2.2 Sampling Plan A. Sampling shall be in accordance with MIL-STD-105, Special Inspection Level S-1, AQL = 6.5 defects per 100 units. Sample units shall be subjected to the following tests:

a. Low temperature

b. High temperature

4.5.2.3 Sampling Plan B. Unless otherwise specified, the sample units selected and tested under 4.5.2.2, for the first lot produced under each contract or order, shall also be subjected to the following tests. The AQL of 6.5 defects per hundred units shall include any defects discovered during the following tests:

a. Vibration

b. Shock

c. Humidity

4.5.2.3.1 Rejection and retest. When the first production lot exhibits any failure under Sampling Plan B, Sampling Plan B shall be required on each succeeding lot until waived in writing by the procuring activity.

4.5.2.3.2 Waiver of Sampling Plan B. Sampling Plan B may be waived if all of the following provisions apply:

a. The contractor has produced this part number and FSN item with consistently satisfactory quality and design.

b. Last delivered items were satisfactory and were delivered within the past 12 months.

c. Waiver is requested by letter giving date and contract number of last delivery.

d. Procuring activity concurs (see 6.2d).

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. Each test set shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

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4.6.2 Room temperature operation. The master-test-ratio-transformer specified in 4.3.4 shall be connected to the transmitter receptable provided on the test set with input power applied as specified in 4.3.3. The master-test-ratio-transformer shall be set to produce the true ratios required to cause the master indicator to register at intervals of 10 percent throughout the scale range. These master indicator readings shall be recorded and known as true values. The master indicator shall then be tested by using the variable0ratio-transformer incorporated within the test set. This shall be accomplished by actuating the master indicator check switch, setting the indicator percent deflection knob at each of the respective value settings, and recording the readings of the master indicator. These readings shall be known as reference values and shall not differ from the true values by more than ±0.5 percent of full scsle. During the above tests the following components of the test set shall perform satisfactorily:

a. Ratio transformer

b. Frequency meter

c. Voltmeter

d. All controls and switches

e. Variable power transformer (shall provide 26 volts output when the input power is varied from 20 32 volts).

4.6.2.1 Cable test. Each cable shall be checked for continuity of wiring.

4.6.3 Low temperature. The test set shall be placed in a temperature chamber and subjected to a temperature of -62°C for a period of 48 hours. The temperature shall then be changed to -40°C for a period of 24 hours. At the conclusion of this exposure period and while at this temperature, the test set shall be subjected to the tests specified in 4.6.2 with the master-test-ratio transmitter at room temperature. The low temperature readings shall not differ from the reference values by more than ±1.25 percent of full scale. After the test set has stabilized at room temperature for a minimum period of 12 hours, it shall meet the tests and tolerances specified in 4.6.2. There shall be no evidence of damage that would adversely affect subsequent operation or accuracy of the test set.

4.6.4 High temperature. The test set shall be placed in a temperature chamber and subjected to a temperature of ±71°C (with an internal relative humidity of not more than 15 percent) for a period of 48 hours. At the end of the 48-hour period, the temperature shall be changed to ±55°C for a period of 4 hours. At the conclusion of this exposure period and while still at this temperature, the test set shall be subjected to the tests specified under 4.6.2. The readings shall not differ respectively from the sequence values by more than ±1.25 percent of full scale. After the test set has returned to room temperature for a minimum of 12 hours, it shall meet the tests and tolerances specified in 4.6.2. There shall be no evidence of damage that would affect subsequent operation or accuracy of the test set.

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4.6.5 Vibration. The test set, with the lid closed and secured, shall be mounted on a vibration test stand and subjected to vibration at frequency varying between 10 and 55 cycles per second at a total excursion of 0.03 ±0.006 inch. The frequency shall be varied uniformly from 10 to 55 cycles per second and returned to 10 cycles per second in approximately 1 minute. The vibration shall be applied in each of the following directions for a period of 30 minutes.

a. Horizontally, parallel to the major horizontal axis of the test set

b. Horizontally, at right angles to the major horizontal axis of the test set.

c. Vertically

Upon completion of the vibration test, the test set shall be subjected to the tests specified in 4.6.2. The readings shall not differ from the reference values by more than 0.5 percent of full scale. No damage shall be noted that would affect subsequent operation or accuracy of test set.

4.6.6 Shock. The test set shall be subjected to impact shocks of 20g with each shock impulse having a time duration of 11 ±1 millisecond in accordance with the test procedure outlined in MIL-STD-202. The shock shall be applied in the following directions:

a. Horizontally parallel to the major horizontal axis (3 shocks in each direction for a total of 6 shocks)

b. Horizontally at right angles to the major horizontal axis (3 shocks in each direction for a total of 6 shocks)

c. Vertically (3 shocks in each direction for a total of 6 shocks)

Upon completion of the vibration test, the test set shall be subjected to the tests specified in 4.6.2. The readings shall not differ from the reference values by more than 0.5 percent of full scale. No damage shall be noted that would adversely affect subsequent operation or accuracy of test set.

* 4.6.7 Humidity. The test set shall be so positioned in the test chamber that it will rest on the 4 feet provided on the side of the case opposite the lid. The lid shall be open. The humidity test shall be conducted in accordance with MIL-STD-810. Within 1 hour after the end of the 240-hour period, the test set shall again be subjected to the tests specified in 4.6.2. The readings shall not differ from the reference values by more than 0.5 percent of full scale. There shall be no evidence of corrosion that would affect subsequent operation or accuracy of test set.

5. PACKAGING

5.1 Preservation and Packaging. Preservation and packaging shall be level A, C or Commercial as specified (see 6.2).

5.1.1 Level A. Each test set shall be preserved and packaged in accordande with Method II of MIL-P-116.

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5.1.2 Level C. The Level C preservation for the test sets shall conform to the MIL-STD-794 requirements for this level.

OR

5.1.3 Commercial. The commercial preservation of the test sets shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or Commercial as specified (see 6.2).

5.2.1 Level A. Test sets preserved and packaged as specified in 5.1.1 shall be packed in overseas type containers conforming to PPP-B-601. Where practable, the exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the container specification.

5.2.2 Level B. Test sets preserved and packaged as specified in 5.1.1 shall be packed in weather resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall be limited to the special requirements table of the specification. Closure shall be in accordance with the closure requirements of the box specification.

5.2.3 Level C. The Level C packing for the test sets shall conform to the MIL-STD-794 requirements for this level.

OR

5.2.4 Commercial. The preserved test sets shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contractor or purchase order (see 6.2) each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with the requirements of MIL-STD-129 or ASTM D 3951, as applicable.

5.4 Palletization. When specified (see 6.2) unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

6. NOTES

* 6.1 Intended use. The test set covered by this specification is intended to be used as a portable field instrument for checking performance characteristics of variable reluctance-type pressure indicators and transmitters. Representative types of equipment which will be checked by the test set are covered by MIL-T-25435, MIL-T-38230, MIL-I-25437, and MIL-I-25438.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial.

c. Reinspection date markings.

d. Whether sampling plan B tests in accordance with 4.5.2.3.2 will be considered.

e. When palletization is required (see 5.4)

* 6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contant irrespective of the marginal notations and relationship to the last previous issue.

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